

Unit 6: Inner products and bilinear forms

Inner products

From coordinates to a scalar product

All vector spaces in the main text are real. An **inner product** on V assigns a number $\langle u, v \rangle$ to each ordered pair, satisfying

$$\langle u, v \rangle = \langle v, u \rangle, \quad \langle au + bw, v \rangle = a \langle u, v \rangle + b \langle w, v \rangle, \quad \langle v, v \rangle > 0 \quad (v \neq 0).$$

Linearity gives $\langle 0, v \rangle = 0$, and symmetry gives linearity in the second argument. Define $\|v\| = \sqrt{\langle v, v \rangle}$. Vectors are orthogonal if their inner product is zero. An orthonormal list consists of mutually orthogonal unit vectors.

The coordinate product on $\mathbb{R}^n$ is one example. Two further examples will keep the abstraction concrete. On real $m \times n$ matrices,

$$\langle A, B \rangle_F = \sum_{i,j} a_{ij} b_{ij} = \text{tr}(A^\top B)$$

is an inner product, with $\|A\|_F^2 = \sum_{i,j} a_{ij}^2$. Indeed the first two axioms follow from finite sums and the third from the sum of squares. This is the **Frobenius inner product**. On the real continuous functions on $[0, 1]$,

$$\langle f, g \rangle = \int_0^1 f(t)g(t) dt$$

is an inner product. Symmetry and linearity follow from integration. If $f \neq 0$, continuity gives an interval of positive length on which $|f|$ is bounded below by a positive number, so $\int_0^1 f^2 > 0$. This argument uses continuity; a value at a single point would not suffice.

Expansion of the axioms gives

$$\|u + v\|^2 = \|u\|^2 + 2 \langle u, v \rangle + \|v\|^2, \quad \|cv\| = |c| \|v\|.$$

Thus Pythagoras holds for orthogonal vectors in any real inner-product space, not only for coordinate vectors.

Proposition 6.1 (Coefficients in an orthogonal list). Nonzero mutually orthogonal vectors $u_1, \dots, u_r$ are independent. If $v = \sum_i c_i u_i$, then

$$c_j = \frac{\langle v, u_j \rangle}{\|u_j\|^2}.$$

For arbitrary v , the vector

$$p = \sum_{i=1}^r \frac{\langle v, u_i \rangle}{\|u_i\|^2} u_i$$

satisfies $v - p \perp u_j$ for every j , hence $v - p$ is perpendicular to their whole span.

Proof. Taking the inner product of $\sum_i c_i u_i$ with u_j leaves $c_j \|u_j\|^2$. Applied to a zero combination, this proves that every coefficient is zero. Applied to v , it gives the formula. For arbitrary v , subtracting the displayed p gives

$$\langle v - p, u_j \rangle = \langle v, u_j \rangle - \frac{\langle v, u_j \rangle}{\|u_j\|^2} \|u_j\|^2 = 0.$$

Linearity gives orthogonality to any combination of the u_j . □

Theorem 6.2 (Projection, best approximation, and Bessel). Let $W = \text{span}\{u_1, \dots, u_r\}$ for a nonzero orthogonal list. The vector p just defined is the unique vector of W with $v - p \perp W$. It is also the unique closest vector of W to v . For every $w \in W$,

$$\|v - w\|^2 = \|v - p\|^2 + \|p - w\|^2.$$

In particular,

$$\sum_{i=1}^r \frac{\langle v, u_i \rangle^2}{\|u_i\|^2} \leq \|v\|^2,$$

with equality exactly when $v \in W$.

Proof. The preceding proposition proves existence of the orthogonal residual. If $p' \in W$ has the same property, $p - p'$ belongs to W and is perpendicular to W ; its inner product with itself is zero, so $p = p'$. Now $v - w = (v - p) + (p - w)$ is an orthogonal sum. Pythagoras proves the distance identity. Its second term vanishes exactly at $w = p$, proving the minimizing assertion with uniqueness. At $w = 0$, the identity gives $\|v\|^2 = \|v - p\|^2 + \|p\|^2$. Expansion in the orthogonal list yields $\|p\|^2 = \sum_i \langle v, u_i \rangle^2 / \|u_i\|^2$. Equality holds precisely when $v = p$. $\square$

For $W = \{0\}$, use the empty list and $p = 0$. For an orthonormal list q_i , the coefficients simplify to $\langle v, q_i \rangle$. When that list is a basis of V , the Bessel inequality is the equality $\|v\|^2 = \sum_i \langle v, q_i \rangle^2$.

Proposition 6.3 (Schwarz and the triangle inequality). For all u, v ,

$$|\langle u, v \rangle| \leq \|u\| \|v\|, \quad \|u + v\| \leq \|u\| + \|v\|.$$

For $v \neq 0$, Schwarz equality means u is a multiple of v ; for $v = 0$, equality always holds.

Proof. For $v \neq 0$, apply the preceding theorem to the one-vector list v , multiply by $\|v\|^2$, and take square roots. Its equality condition gives the claim. If $v = 0$, both sides of Schwarz are zero. Then

$$\|u + v\|^2 = \|u\|^2 + 2\langle u, v \rangle + \|v\|^2 \leq (\|u\| + \|v\|)^2.$$

Both quantities whose squares are compared are nonnegative. $\square$

Example 6.1 (A weighted geometry). Let

$$G = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}, \quad \langle x, y \rangle_G = x^\top G y.$$

Symmetry of G gives symmetry of this pairing, and products give bilinearity. Its positivity follows by completing the square:

$$x^\top G x = 3x_1^2 - 2x_1x_2 + 2x_2^2 = 3(x_1 - x_2/3)^2 + \frac{5}{3}x_2^2 > 0 \quad (x \neq 0).$$

Such a symmetric matrix is called **positive definite**. In this geometry $\langle e_1, e_2 \rangle_G = -1$, so the coordinate axes are not orthogonal. Projection of e_1 onto $\text{span}\{e_2\}$ gives

$$p = \frac{-1}{2}e_2, \quad r = e_1 - p = (1, 1/2)^\top.$$

Check $\langle r, e_2 \rangle_G = -1 + 2(1/2) = 0$. The squared norms are $\|e_1\|_G^2 = 3$, $\|p\|_G^2 = 1/2$, $\|r\|_G^2 = 5/2$. For a general candidate te_2 ,

$$\|e_1 - te_2\|_G^2 = 3 + 2t + 2t^2 = \frac{5}{2} + 2(t + 1/2)^2.$$

The minimum and its unique location agree with the projection formula.

Orthogonal bases and complements

Theorem 6.4 (Gram–Schmidt). For an independent list $v_1, \dots, v_k$, define

$$u_1 = v_1, \quad u_j = v_j - \sum_{i < j} \frac{\langle v_j, u_i \rangle}{\|u_i\|^2} u_i.$$

Every u_j is nonzero, the list is orthogonal, and $\text{span}\{u_1, \dots, u_j\} = \text{span}\{v_1, \dots, v_j\}$ at every stage. Thus $q_j = u_j / \|u_j\|$ is an orthonormal basis of the original span.

Proof. Independence gives $u_1 = v_1 \neq 0$. Suppose the conclusions hold before stage j . The projection calculation makes u_j perpendicular to every earlier u_i . If $u_j = 0$, then v_j is a combination of earlier u_i , and hence of earlier v_i , contrary to independence. The defining formula puts u_j in the span of v_j and earlier vectors; rearranging puts v_j in the span of u_j and earlier vectors. These two inclusions prove equality of the new spans. Induction proves all claims. Normalizing preserves orthogonality and span. $\square$

Proposition 6.5 (Extension and decomposition). An orthonormal list in a finite-dimensional inner-product space extends to an orthonormal basis. If W is a subspace of such a space V , define

$$W^\perp = \{v : \langle v, w \rangle = 0 \text{ for all } w \in W\}.$$

Then $W^\perp$ is a subspace, and

$$V = W \oplus W^\perp, \quad \dim W + \dim W^\perp = \dim V, \quad (W^\perp)^\perp = W.$$

Here the direct sum means every vector has a unique decomposition into one component in each subspace.

Proof. An orthonormal list is independent. Extend it to a basis and apply Gram–Schmidt in that order; its initial orthonormal vectors are unchanged. The remaining vectors normalize to give the claimed extension.

The defining equations of $W^\perp$ are preserved by linear combinations. Choose an orthonormal basis $q_1, \dots, q_r$ of W and extend it to $q_1, \dots, q_n$ of V . In $v = \sum_i \langle v, q_i \rangle q_i$, the first r terms belong to W , and the rest are perpendicular to W . Conversely, if $v \perp W$, its first r coefficients are zero. Thus $q_{r+1}, \dots, q_n$ form a basis of $W^\perp$. This proves existence, the dimension formula, and the double-complement identity. If a vector belongs to both subspaces, it is perpendicular to itself and is zero. Subtracting two decompositions therefore proves uniqueness. Empty lists include the cases $W = 0$ and $W = V$. $\square$

Example 6.2 (A plane: basis, orthogonalization, and projection). Consider $W = \{(x, y, z)^\top : 3x - 2y + z = 0\}$. Because $z = -3x + 2y$, this space has dimension two. The vectors $a = (1, 2, 1)^\top$, $b = (3, 5, 1)^\top$ lie in W and are independent: the first and third coordinates of $\alpha a + \beta b = 0$ give $\alpha + 3\beta = 0$, $\alpha + \beta = 0$, hence $\alpha = \beta = 0$. Now

$$a \cdot a = 6, \quad a \cdot b = 14, \quad b - \frac{14}{6}a = \frac{1}{3}(2, 1, -4)^\top.$$

With $c = (2, 1, -4)^\top$, we have $a \cdot c = 0$, $c \cdot c = 21$. Hence $q_1 = a/\sqrt{6}$, $q_2 = c/\sqrt{21}$ is an orthonormal basis of W . Its complement has unit basis $q_3 = (3, -2, 1)^\top/\sqrt{14}$.

For $v = e_3$, the coefficients in the plane basis are $v \cdot q_1 = 1/\sqrt{6}$, $v \cdot q_2 = -4/\sqrt{21}$. Thus

$$p = \frac{1}{6}a - \frac{4}{21}c = \frac{1}{14}(-3, 2, 13)^\top, \quad r = e_3 - p = \frac{1}{14}(3, -2, 1)^\top.$$

Directly, $3p_1 - 2p_2 + p_3 = 0$. Also $\|r\|^2 = 1/14$, $\|p\|^2 = 13/14$; their sum is $\|e_3\|^2$. The closest-point identity is $\|e_3 - w\|^2 = 1/14 + \|p - w\|^2$ for every $w \in W$.

Proposition 6.6 (Projection matrices). If Q has an orthonormal basis of $W \subseteq \mathbb{R}^n$ as columns, then

$$P_W = QQ^\top, \quad P_W^\top = P_W, \quad P_W^2 = P_W, \quad \text{im } P_W = W, \quad \ker P_W = W^\perp.$$

Also $I - P_W = P_{W^\perp}$. Conversely, a real symmetric idempotent matrix is the orthogonal projection onto its image.

Proof. The coefficient formula is $P_W x = \sum_j q_j(q_j^\top x) = QQ^\top x$. Since $Q^\top Q = I$, transposing and squaring give the identities. The decomposition proves the image, kernel and complementary formula. Conversely, for $P^\top = P = P^2$, Px is in its image and

$$\langle x - Px, Py \rangle = x^\top (P - P^\top P)y = 0.$$

Thus the decomposition relative to the image has component Px , as claimed. For the zero space the projector is zero. $\square$

For the preceding plane, the formulas from its plane basis and its normal $n = (3, -2, 1)^\top$ agree:

$$\frac{aa^\top}{6} + \frac{cc^\top}{21} = \frac{1}{14} \begin{pmatrix} 5 & 6 & -3 \\ 6 & 10 & 2 \\ -3 & 2 & 13 \end{pmatrix} = I - \frac{nn^\top}{14}.$$

The equality can be checked entry by entry; for example the $(1, 2)$ entry is $2/6 + 2/21 = 6/14$.

For a matrix A , $Ax = 0$ means x is perpendicular to every row. Consequently

$$\ker A = \text{row}(A)^\perp, \quad (\ker A)^\perp = \text{row}(A).$$

These identities and rank–nullity give another proof that row rank equals column rank: both equal $n - \dim \ker A$.

Bilinear maps and matrices

A map $g : U \times V \rightarrow W$ is **bilinear** if it is linear in each argument with the other fixed. It need not be linear in the pair: for example $g(u, v) = uv$ on $\mathbb{R} \times \mathbb{R}$ is bilinear, but $g(2u, 2v) = 4uv$, not generally $2uv$. A scalar-valued bilinear map is a **bilinear form**. Matrix multiplication is bilinear in its two matrix arguments; the scalar-valued coordinate example is

$$g_A(x, y) = x^\top Ay = \sum_{i=1}^m \sum_{j=1}^n a_{ij} x_i y_j, \quad A \in \mathbb{R}^{m \times n}, \quad x \in \mathbb{R}^m, \quad y \in \mathbb{R}^n.$$

Theorem 6.7 (Representation of a bilinear form). Given ordered bases $B = (b_1, \dots, b_m)$ of U and $C = (c_1, \dots, c_n)$ of V , every bilinear form g has a unique matrix $A \in \mathbb{R}^{m \times n}$ such that

$$g(u, v) = [u]_B^\top A [v]_C, \quad a_{ij} = g(b_i, c_j).$$

This correspondence is a linear bijection between matrices and bilinear forms. In particular the space of these forms has dimension mn .

Proof. Write $u = \sum_i x_i b_i$, $v = \sum_j y_j c_j$. Linearity in the first argument and then in the second gives $g(u, v) = \sum_{i,j} x_i y_j g(b_i, c_j)$, proving existence. If two matrices give all the same values, substitute $u = b_i, v = c_j$ to recover each entry, proving uniqueness. Conversely the coordinate formula for any A is bilinear. Finally $g_{A+D} = g_A + g_D$, $g_{aA} = ag_A$, by the product laws. Thus the correspondence is linear and bijective; the matrix units give mn independent coordinate directions. $\square$

Proposition 6.8 (Symmetry and a change of basis). On one space with the same basis in both positions, g_A is symmetric if and only if $A = A^\top$. A symmetric bilinear form is an inner product if and only if $x^\top Ax > 0$ for every nonzero coordinate column x . If S converts new first-argument coordinates to old ones, and T does the same for the second argument, the new matrix is

$$A_{\text{new}} = S^\top A_{\text{old}} T.$$

In particular the Gram matrix of an inner product changes by $G_{\text{new}} = S^\top G_{\text{old}} S$.

Proof. The scalar identity $y^\top Ax = x^\top A^\top y$ proves the symmetry claim by uniqueness of the representing matrix. Bilinearity is already known, so precisely positivity remains for an inner product. If the new coordinate columns are x', y' , then the old ones are Sx', Ty' ; substitution gives $g = x'^\top S^\top A T y'$, proving the transformation rule. For an inner product, $G_{ij} = \langle b_i, b_j \rangle$; it is symmetric, and

$$x^\top Gx = \left\| \sum_i x_i b_i \right\|^2 > 0 \quad (x \neq 0),$$

because the basis is independent. □

Example 6.3 (Reading and changing a bilinear form). For $A = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$,

$$g_A(x, y) = x_1 y_1 + 2x_1 y_2 + 3x_2 y_1 - x_2 y_2.$$

For example $g_A(e_1, e_2) = 2$, whereas $g_A(e_2, e_1) = 3$; this form is not symmetric. Take the new basis $b_1 = (1, 1)^\top, b_2 = (1, -1)^\top$, with $S = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Then

$$AS = \begin{pmatrix} 3 & -1 \\ 2 & 4 \end{pmatrix}, \quad S^\top AS = \begin{pmatrix} 5 & 3 \\ 1 & -5 \end{pmatrix}.$$

The $(1, 2)$ entry is checked directly: $g_A(b_1, b_2) = (1, 1)(-1, 4)^\top = 3$. For the ordinary dot product the same basis has Gram matrix $S^\top S = 2I$. Thus the basis is orthogonal with squared lengths two, and its coordinate norm formula is $2x_1'^2 + 2x_2'^2$.

Calculus connection. For a twice continuously differentiable real function f near a point, its Hessian $H = (\partial^2 f / \partial x_i \partial x_j)$ represents the bilinear form $(u, v) \mapsto u^\top H v$. Equality of mixed partials under this regularity assumption makes it symmetric. Symmetry alone does not imply positivity: $f(x_1, x_2) = x_1^2 - x_2^2$ has Hessian $\text{diag}(2, -2)$.

Exercises and solutions

All spaces and scalars are real. **Suggested practice:** 6.1, 6.6, 6.8, 6.12, 6.15, 6.16, 6.18, 6.19, 6.21, 6.23, 6.26, 6.31.

Inner products and their norms

Exercise 6.1. Verify the inner-product axioms for (a) $\mathbb{R}^m$ with $\sum_i x_i y_i$; (b) real $m \times n$ matrices with $\text{tr}(A^\top B)$.

Solution. In (a), symmetry and linearity follow term by term; $\sum_i x_i^2$ is nonnegative and vanishes precisely when every coordinate is zero. In (b), $\text{tr}(A^\top B) = \sum_{i,j} a_{ij} b_{ij}$, so the same argument applies to the entries.

Exercise 6.2. In a real inner-product space prove $|\langle x, y \rangle| \leq \|x\| \|y\|$. Determine exactly when equality holds, including zero vectors.

Solution. If $y = 0$, both sides vanish. For $y \neq 0$, put $r = x - \langle x, y \rangle y / \|y\|^2$. Then

$$0 \leq \|r\|^2 = \|x\|^2 - \frac{\langle x, y \rangle^2}{\|y\|^2}.$$

Multiplication by $\|y\|^2$ proves the inequality. Equality holds exactly when $r = 0$, or x is a scalar multiple of y . Including the case $y = 0$, the condition is that the pair is linearly dependent.

Exercise 6.3. Prove $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$ and explain its name, the parallelogram identity.

Solution. The two squares expand to $\|x\|^2 \pm 2\langle x, y \rangle + \|y\|^2$; adding cancels the mixed terms. A parallelogram with adjacent side vectors x, y has diagonals $x + y, x - y$. Thus the sum of the squared diagonal lengths equals the sum of the squared lengths of its four sides.

Exercise 6.4. A norm satisfies $\nu(ax) = |a|\nu(x)$, positivity with equality only at zero, and $\nu(x + y) \leq \nu(x) + \nu(y)$. Prove that $\nu(x) = \sqrt{\langle x, x \rangle}$ is a norm. Include the Frobenius norm as a special case.

Solution. Bilinearity gives $\langle ax, ax \rangle = a^2 \langle x, x \rangle$, proving homogeneity. Positive definiteness gives positivity and its equality condition. Schwarz gives

$$\|x + y\|^2 = \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \leq (\|x\| + \|y\|)^2.$$

Taking nonnegative square roots proves the triangle inequality. For matrices the product is $\text{tr}(A^\top B)$, so the norm is $\sqrt{\sum_{i,j} a_{ij}^2}$ and all three conclusions apply.

Exercise 6.5. Prove that a real inner product is recovered from its norm by

$$\langle x, y \rangle = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2).$$

Solution. Expanding the two squares and subtracting cancels the two pure-square terms and leaves $4\langle x, y \rangle$. In particular two inner products inducing the same norm must agree on every pair.

Exercise 6.6. For $m \geq 2$, let $\nu(x) = \sum_{i=1}^m |x_i|$ on $\mathbb{R}^m$. Prove that ν is a norm but is not induced by an inner product.

Solution. Absolute values give $\nu(ax) = |a|\nu(x)$ and $\nu(x+y) \leq \sum_i (|x_i| + |y_i|) = \nu(x) + \nu(y)$. Every summand is nonnegative; their sum is zero exactly when $x = 0$. For distinct coordinate vectors e_1, e_2 , however,

$$\nu(e_1 + e_2)^2 + \nu(e_1 - e_2)^2 = 8, \quad 2\nu(e_1)^2 + 2\nu(e_2)^2 = 4.$$

The parallelogram identity fails, so no inner product induces this norm.

Exercise 6.7. For $d(x, y) = \|x - y\|$, prove (a) $d(x, x) = 0$; (b) $d(x, y) > 0$ for $x \neq y$; (c) symmetry; (d) $d(x, y) \leq d(x, z) + d(z, y)$. Interpret these in $\mathbb{R}^2$.

Solution. The first two assertions are norm positivity. Symmetry follows from $\|-(x - y)\| = \|x - y\|$. Finally $x - y = (x - z) + (z - y)$, so the norm triangle inequality proves (d). In the plane, coincident points have distance zero and distinct points have positive distance; reversing a route does not change its length; and the direct segment is no longer than the two-segment route through z .

Exercise 6.8. Suppose $\|x_k - x\| \rightarrow 0$ and $\|y_k - y\| \rightarrow 0$ in a real inner-product space. Prove convergence of the norms and of $\langle x_k, y_k \rangle$ to $\langle x, y \rangle$.

Solution. The triangle inequality in both directions gives $|\|x_k\| - \|x\|| \leq \|x_k - x\| \rightarrow 0$, and similarly for y_k . In particular $\|x_k\|$ is bounded. By bilinearity and Schwarz,

$$|\langle x_k, y_k \rangle - \langle x, y \rangle| \leq \|x_k\| \|y_k - y\| + \|x_k - x\| \|y\| \rightarrow 0.$$

Orthogonality and construction

Exercise 6.9. Show that zero is orthogonal to every vector. Find every vector in $\mathbb{R}^3$ orthogonal to $(2, 2, -1)^\top$.

Solution. Linearity gives $\langle 0, v \rangle = \langle 0 + 0, v \rangle = 2\langle 0, v \rangle$, hence zero. The coordinate condition is $2x + 2y - z = 0$; its complete solution is $(s, t, 2s + 2t)^\top = s(1, 0, 2)^\top + t(0, 1, 2)^\top$, with $s, t \in \mathbb{R}$.

Exercise 6.10. In a real inner-product space show that $x \perp y$ implies $\|x + y\|^2 = \|x\|^2 + \|y\|^2$. Does the converse hold?

Solution. Expansion gives $\|x + y\|^2 - \|x\|^2 - \|y\|^2 = 2\langle x, y \rangle$. Thus each condition is equivalent to the other over the real field.

Exercise 6.11. Show that the columns of I_n are orthonormal and independent. Then prove that every orthonormal list is independent.

Solution. For coordinate columns, $e_i^\top e_j$ equals one for $i = j$ and zero otherwise, and $\sum_i c_i e_i = (c_1, \dots, c_n)^\top$ vanishes only when all coefficients vanish. More generally, taking the inner product of $\sum_i c_i q_i = 0$ with q_j gives $c_j = 0$ for every j .

Exercise 6.12. Two subspaces are orthogonal when every vector of either is orthogonal to every vector of the other. Explain why a room's front and side walls, viewed as planes through a corner, are not orthogonal subspaces. Show also that the zero subspace is orthogonal to every subspace.

Solution. A nonzero vector along the common vertical edge belongs to both planes. It has positive inner product with itself, contradicting subspace orthogonality. The zero subspace contains only zero, whose inner product with every vector is zero.

Exercise 6.13. For a subspace W of an inner-product space V , determine $\{0\}^\perp$, prove that $W^\perp$ is a subspace, and show that $W \cap W^\perp = \{0\}$.

Solution. Every vector is perpendicular to zero, so $\{0\}^\perp = V$. If $x, y \in W^\perp$, then for every $w \in W$, $\langle ax + by, w \rangle = a \langle x, w \rangle + b \langle y, w \rangle = 0$; zero also belongs to the set. Finally a vector in the intersection is perpendicular to itself, so positive definiteness forces it to be zero.

Exercise 6.14. Prove that every finite-dimensional real inner-product space has an orthonormal basis.

Solution. For the zero space use the empty list. Otherwise take a basis $v_1, \dots, v_n$, and recursively set $u_j = v_j - \sum_{i < j} \langle v_j, u_i \rangle u_i / \|u_i\|^2$. Inductively the previously constructed u_i are nonzero and mutually orthogonal, and their span equals that of $v_1, \dots, v_{j-1}$. Taking inner products proves $u_j \perp u_i$. If $u_j = 0$, then v_j is in the preceding span, contradicting independence. The formula and its rearrangement show equality of the two spans at stage j . Thus all u_j exist, and $u_j / \|u_j\|$ form the required orthonormal basis.

Exercise 6.15. Apply Gram–Schmidt, in the given order, to obtain orthonormal bases for the spans of these lists:

- (a) $(1, 1, -1)^\top, (1, 0, 1)^\top$;
- (b) $(2, 1, 1)^\top, (1, 3, -1)^\top$;
- (c) $(1, 2, 1, 0)^\top, (1, 2, 3, 1)^\top$;
- (d) $(1, 1, 0, 0)^\top, (1, -1, 1, 1)^\top, (-1, 0, 2, 1)^\top$.

Solution. (a) The vectors are already orthogonal, with squared norms 3, 2; divide them by $\sqrt{3}, \sqrt{2}$. (b) The first squared norm is 6, and the mixed product is 4. The residual is $(1, 3, -1)^\top - \frac{2}{3}(2, 1, 1)^\top = \frac{1}{3}(-1, 7, -5)^\top$. An orthonormal basis is $(2, 1, 1)^\top / \sqrt{6}, (-1, 7, -5)^\top / \sqrt{75}$. (c) The mixed product is 8, and the first squared norm is 6. Subtracting $\frac{4}{3}(1, 2, 1, 0)^\top$ leaves $\frac{1}{3}(-1, -2, 5, 3)^\top$. Divide the directions $(1, 2, 1, 0)^\top, (-1, -2, 5, 3)^\top$ by $\sqrt{6}, \sqrt{39}$. (d) The first two vectors are orthogonal, with squared norms 2, 4. Their inner products with the third are $-1, 2$, respectively, so its residual is

$$(-1, 0, 2, 1)^\top + \frac{1}{2}(1, 1, 0, 0)^\top - \frac{1}{2}(1, -1, 1, 1)^\top = \frac{1}{2}(-2, 2, 3, 1)^\top.$$

The required basis consists of the first two vectors divided by $\sqrt{2}, 2$, followed by $(-2, 2, 3, 1)^\top / \sqrt{18}$.

Exercise 6.16. Find an orthogonal basis for each solution space:

- (a) $2x + y - z = 0, \quad y + z = 0$;
- (b) $x - y + z = 0$;
- (c) $4x + 7y - \pi z = 0, \quad 2x - y + z = 0$;
- (d) $x + y + z = 0, \quad x - y = 0, \quad y + z = 0$.

Solution. (a) The equations give $(x, y, z) = t(1, -1, 1)$; the single displayed nonzero vector is an orthogonal basis. (b) Write $(x, y, z) = s(1, 1, 0) + t(-1, 0, 1)$. Subtracting the projection of the second direction onto the first leaves $(-1/2, 1/2, 1)$, so $(1, 1, 0)^\top, (-1, 1, 2)^\top$ is an orthogonal basis. (c) Substitution $y = 2x + z$ gives $18x + (7 - \pi)z = 0$. Thus the entire solution space is spanned by $(\pi - 7, 2\pi + 4, 18)^\top$, which is nonzero. (d) The second and third equations give $x = y = -z$; the first forces $x = 0$. The zero space has the empty orthogonal basis.

Exercise 6.17. For $\langle f, g \rangle = \int_0^1 fg$, apply Gram–Schmidt to t, t^2 and obtain an orthonormal basis of their span.

Solution. The first squared norm is $1/3$, and $\langle t^2, t \rangle = 1/4$. Hence the residual is $u = t^2 - \frac{3}{4}t$, with

$$\|u\|^2 = \frac{1}{5} - \frac{3}{2} \frac{1}{4} + \frac{9}{16} \frac{1}{3} = \frac{1}{80}.$$

Normalize to obtain $\sqrt{3}t$ and $\sqrt{5}(4t^2 - 3t)$. The nonzero residual also verifies independence of the original pair.

Exercise 6.18. For $\langle f, g \rangle = \int_0^1 fg$, apply Gram–Schmidt to $1, t, t^2$ and give an orthonormal basis of their span.

Solution. The first vector is 1 , already normalized. The next residual is $u_2 = t - \frac{1}{2}$, with squared norm $1/12$. For t^2 , the products with $1, u_2$ are $1/3, 1/12$, so

$$u_3 = t^2 - \frac{1}{3} - u_2 = t^2 - t + \frac{1}{6}, \quad \|u_3\|^2 = \frac{1}{5} - \frac{1}{2} + \frac{4}{9} - \frac{1}{6} + \frac{1}{36} = \frac{1}{180}.$$

Thus an orthonormal basis is $1, \sqrt{3}(2t - 1), \sqrt{5}(6t^2 - 6t + 1)$.

Exercise 6.19. Let $A_1, \dots, A_k$ be matrices in a real inner-product space, possibly dependent. Show that coefficients can be chosen so that $B_j = A_j - \sum_{i < j} c_{ij} B_i$ are mutually orthogonal. Show that $c_{ij} = \langle A_j, B_i \rangle / \|B_i\|^2$ is forced when $B_i \neq 0$, and determine how many B_i are nonzero. Describe how to obtain an orthonormal basis for the original span.

Solution. Construct successively. For each earlier nonzero B_i , choose the displayed coefficient, and choose zero when $B_i = 0$. Orthogonality of the previous list gives $\langle B_j, B_i \rangle = \langle A_j, B_i \rangle - c_{ij} \|B_i\|^2 = 0$ for nonzero B_i ; for zero B_i it is automatic. The same equation proves uniqueness of each coefficient attached to a nonzero vector; coefficients attached to zero are arbitrary. The triangular formulas in both directions show $\text{span}(B_1, \dots, B_j) = \text{span}(A_1, \dots, A_j)$. The nonzero B_i are orthogonal and hence independent. They therefore form a basis of the original span, and their number is its dimension. Discard zero vectors and normalize the remaining ones to get the requested orthonormal basis.

Exercise 6.20. Let A be a real $m \times k$ matrix of rank r . Show that $A = QR_1$, where Q has r orthonormal columns and R_1 consists of the nonzero rows of an upper triangular $k \times k$ matrix R . Require R to have r positive diagonal entries and $k - r$ zero rows.

Solution. Apply the preceding construction to the columns a_j , obtaining orthogonal residuals b_j . For every nonzero b_i , set $q_i = b_i / \|b_i\|$, keeping its original index for now, and set $R_{ij} = \langle q_i, a_j \rangle$. For zero b_i , set row i of R equal to zero. A nonzero b_i is perpendicular to the span of $a_1, \dots, a_{i-1}$, so $R_{ij} = 0$ for $j < i$; further $R_{ii} = \|b_i\| > 0$. Each a_j lies in the span of the nonzero residuals through index j , so its orthogonal expansion is $a_j = \sum_{i \leq j, b_i \neq 0} q_i R_{ij}$. Assemble the r nonzero normalized residuals into Q , and retain exactly their rows in R_1 . These column identities give $A = QR_1$ and $Q^\top Q = I_r$. For $r = 0$, use an empty Q, R_1 and $R = 0$.

Exercise 6.21. On the space V of real symmetric $n \times n$ matrices, $n \geq 1$, prove that $\langle A, B \rangle = \text{tr}(AB)$ is an inner product. If $W = \{A \in V : \text{tr } A = 0\}$, prove $\dim W^\perp = 1$.

Solution. For symmetric matrices $\text{tr}(AB) = \text{tr}(A^\top B) = \sum_{i,j} a_{ij} b_{ij}$, so this is the restriction of the Frobenius inner product. Also $\langle I, A \rangle = \text{tr } A$, hence $\text{span}(I) \subseteq W^\perp$. If $B \in W^\perp$, let $C = B - (\text{tr } B/n)I$. Then $C \in W$, and $0 = \langle B, C \rangle = \langle C, C \rangle + (\text{tr } B/n) \text{tr } C = \|C\|^2$. Thus $B = (\text{tr } B/n)I$, so $W^\perp = \text{span}(I)$, of dimension one.

Exercise 6.22. For a subspace W of a finite-dimensional inner-product space V , prove $\dim W + \dim W^\perp = \dim V$.

Solution. Take an orthonormal basis $q_1, \dots, q_k$ of W , extend it to a basis of V , and Gram–Schmidt the added vectors. This leaves the first k vectors unchanged and gives an orthonormal basis $q_1, \dots, q_n$ of V . The last $n - k$ vectors are perpendicular to W . Conversely, if $v \perp W$, its coefficients along the first k basis vectors vanish, so it is in the span of the last $n - k$. They form a basis of $W^\perp$, proving the formula, including $k = 0, n$.

Forms, matrices, and functions

Exercise 6.23. For symmetric A , let $g(x, y) = x^\top Ay$.

- (a) Prove symmetry and linearity in each argument.
- (b) Give a 1×1 and a nonzero 2×2 matrix for which positivity fails.
- (c) Give a symmetric positive definite 2×2 example.
- (d) If $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$, $a > 0$, prove positivity is equivalent to $ad - b^2 > 0$.
- (e) If $a < 0$, show that positivity fails.

Solution. (a) Matrix multiplication is linear in each vector and $x^\top Ay = (x^\top Ay)^\top = y^\top A^\top x = y^\top Ax$. (b) Take (-1) and $\text{diag}(1, 0)$; the former is negative on 1, and the latter vanishes on nonzero e_2 . (c) I_2 works because $x_1^2 + x_2^2 > 0$ for nonzero x . (d) Completing the square gives

$$ax^2 + 2bxy + dy^2 = a(x + by/a)^2 + \frac{ad - b^2}{a}y^2.$$

If both coefficients are positive, the expression vanishes only at $(0, 0)$. Conversely set $(x, y) = (-b/a, 1)$; positive definiteness forces $(ad - b^2)/a > 0$. (e) Evaluation at e_1 gives $a < 0$, contradicting positivity.

Exercise 6.24. Determine which matrices are positive definite:

$$\begin{array}{lll} \text{(a)} \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} & \text{(b)} \begin{pmatrix} -2 & 1 \\ 1 & 5 \end{pmatrix} & \text{(c)} \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix} \\ \text{(d)} \begin{pmatrix} 4 & 4 \\ 4 & 1 \end{pmatrix} & \text{(e)} \begin{pmatrix} 4 & 1 \\ 1 & 10 \end{pmatrix} & \text{(f)} \begin{pmatrix} 4 & -1 \\ -1 & 10 \end{pmatrix}. \end{array}$$

Solution. In (a), (c), (e), (f), the first diagonal entries are positive and the values $ad - b^2$ are respectively 5, 7, 39, 39; all four are positive definite by the preceding square completion. In (b), $e_1^\top Ae_1 = -2$. In (d), evaluating at $(-1, 1)^\top$ gives $4 - 8 + 1 = -3$. Thus (b) and (d) fail positivity.

Exercise 6.25. For a real square matrix A , prove that $g_A(x, y) = x^\top Ay$ is symmetric in its arguments if and only if $A = A^\top$.

Solution. If $A = A^\top$, transposing the scalar $x^\top Ay$ gives $x^\top Ay = y^\top A^\top x = y^\top Ax$. Conversely, symmetry on (e_i, e_j) gives $a_{ij} = g_A(e_i, e_j) = g_A(e_j, e_i) = a_{ji}$ for every i, j . Thus $A = A^\top$. This assertion alone imposes no positivity condition.

Exercise 6.26. Write $x^\top Ay$ fully in coordinates for each matrix:

$$\begin{array}{lll} \text{(a)} \begin{pmatrix} 2 & -3 \\ 4 & 1 \end{pmatrix} & \text{(b)} \begin{pmatrix} 4 & 1 \\ -2 & 5 \end{pmatrix} & \text{(c)} \begin{pmatrix} -5 & 2 \\ \pi & 7 \end{pmatrix} \\ \text{(d)} \begin{pmatrix} 1 & 2 & -1 \\ -3 & 1 & 4 \\ 2 & 5 & -1 \end{pmatrix} & \text{(e)} \begin{pmatrix} -4 & 2 & 1 \\ 3 & 1 & 1 \\ 2 & 5 & 7 \end{pmatrix} & \text{(f)} \begin{pmatrix} -\frac{1}{2} & 2 & -5 \\ 1 & \frac{2}{3} & 4 \\ -1 & 0 & 3 \end{pmatrix}. \end{array}$$

Solution. Use $x^\top Ay = \sum_{i,j} a_{ij}x_iy_j$. The answers are

$$\begin{array}{ll} \text{(a)} & 2x_1y_1 - 3x_1y_2 + 4x_2y_1 + x_2y_2, \\ \text{(b)} & 4x_1y_1 + x_1y_2 - 2x_2y_1 + 5x_2y_2, \\ \text{(c)} & -5x_1y_1 + 2x_1y_2 + \pi x_2y_1 + 7x_2y_2, \\ \text{(d)} & x_1y_1 + 2x_1y_2 - x_1y_3 - 3x_2y_1 + x_2y_2 + 4x_2y_3 \\ & \quad + 2x_3y_1 + 5x_3y_2 - x_3y_3, \\ \text{(e)} & -4x_1y_1 + 2x_1y_2 + x_1y_3 + 3x_2y_1 + x_2y_2 + x_2y_3 \\ & \quad + 2x_3y_1 + 5x_3y_2 + 7x_3y_3, \\ \text{(f)} & -\frac{1}{2}x_1y_1 + 2x_1y_2 - 5x_1y_3 + x_2y_1 + \frac{2}{3}x_2y_2 + 4x_2y_3 \\ & \quad - x_3y_1 + 3x_3y_3. \end{array}$$

Exercise 6.27. On the continuous real functions on $[0, 2\pi]$, use $\langle f, g \rangle = \int_0^{2\pi} fg$. For $g_n(t) = \cos(nt)$, $n \geq 0$, and $h_m(t) = \sin(mt)$, $m \geq 1$, compute their norms. Show that $g_n \perp g_m$ when $n \neq m$ and that $g_n \perp h_m$ always.

Solution. For nonzero integer k , the integrals of $\cos kt$ and $\sin kt$ over a full period vanish. Use $2 \cos nt \cos mt = \cos((n-m)t) + \cos((n+m)t)$ and $2 \cos nt \sin mt = \sin((m+n)t) + \sin((m-n)t)$. The first integral is zero for $n \neq m$, and the second is always zero, including the zero-frequency term. Also $\cos^2 nt = (1 + \cos 2nt)/2$ for $n \geq 1$ and $\sin^2 mt = (1 - \cos 2mt)/2$. Hence $\|g_0\| = \sqrt{2\pi}$, and $\|g_n\| = \|h_m\| = \sqrt{\pi}$ for positive indices.

Exercise 6.28. With the preceding product, compute the inner products of $f(t) = t$ with g_n, h_n , and its Fourier coefficients $\langle f, g_n \rangle / \|g_n\|^2$, $\langle f, h_n \rangle / \|h_n\|^2$.

Solution. For $n = 0$, $\langle t, 1 \rangle = 2\pi^2$, giving coefficient π . For $n \geq 1$, integration by parts gives

$$\begin{aligned} \int_0^{2\pi} t \cos nt \, dt &= \left[\frac{t \sin nt}{n} + \frac{\cos nt}{n^2} \right]_0^{2\pi} = 0, \\ \int_0^{2\pi} t \sin nt \, dt &= \left[-\frac{t \cos nt}{n} + \frac{\sin nt}{n^2} \right]_0^{2\pi} = -\frac{2\pi}{n}. \end{aligned}$$

Dividing by the squared norms π gives coefficients $0, -2/n$.

Exercise 6.29. Compute the same inner products and Fourier coefficients for $f(t) = t^2$.

Solution. The constant inner product is $\int_0^{2\pi} t^2 dt = 8\pi^3/3$, giving coefficient $4\pi^2/3$. For $n \geq 1$, integration by parts, using the two integrals in the preceding solution, gives

$$\begin{aligned} \int_0^{2\pi} t^2 \cos nt \, dt &= -\frac{2}{n} \int_0^{2\pi} t \sin nt \, dt = \frac{4\pi}{n^2}, \\ \int_0^{2\pi} t^2 \sin nt \, dt &= -\frac{4\pi^2}{n} + \frac{2}{n} \int_0^{2\pi} t \cos nt \, dt = -\frac{4\pi^2}{n}. \end{aligned}$$

The nonconstant coefficients are therefore $4/n^2$ and $-4\pi/n$.

Exercise 6.30. For the same integral product on $[0, 2\pi]$, find the norms of $f(t) = t$ and $g(t) = t^2$.

Solution. By definition, $\|f\|^2 = \int_0^{2\pi} t^2 dt = 8\pi^3/3$ and $\|g\|^2 = \int_0^{2\pi} t^4 dt = 32\pi^5/5$. The norms are the positive square roots of these quantities.

Exercise 6.31. Let $q_1, \dots, q_m$ be orthonormal in a finite-dimensional real inner-product space V . Suppose $\|v\|^2 = \sum_i \langle v, q_i \rangle^2$ for every $v \in V$. Prove that the list is a basis of V .

Solution. Orthogonality proves independence. For any v , set $p = \sum_i \langle v, q_i \rangle q_i$. Direct expansion gives

$$\|v - p\|^2 = \|v\|^2 - 2 \sum_i \langle v, q_i \rangle^2 + \sum_i \langle v, q_i \rangle^2 = 0.$$

Thus $v = p$, proving that the list spans V . It is therefore a basis.